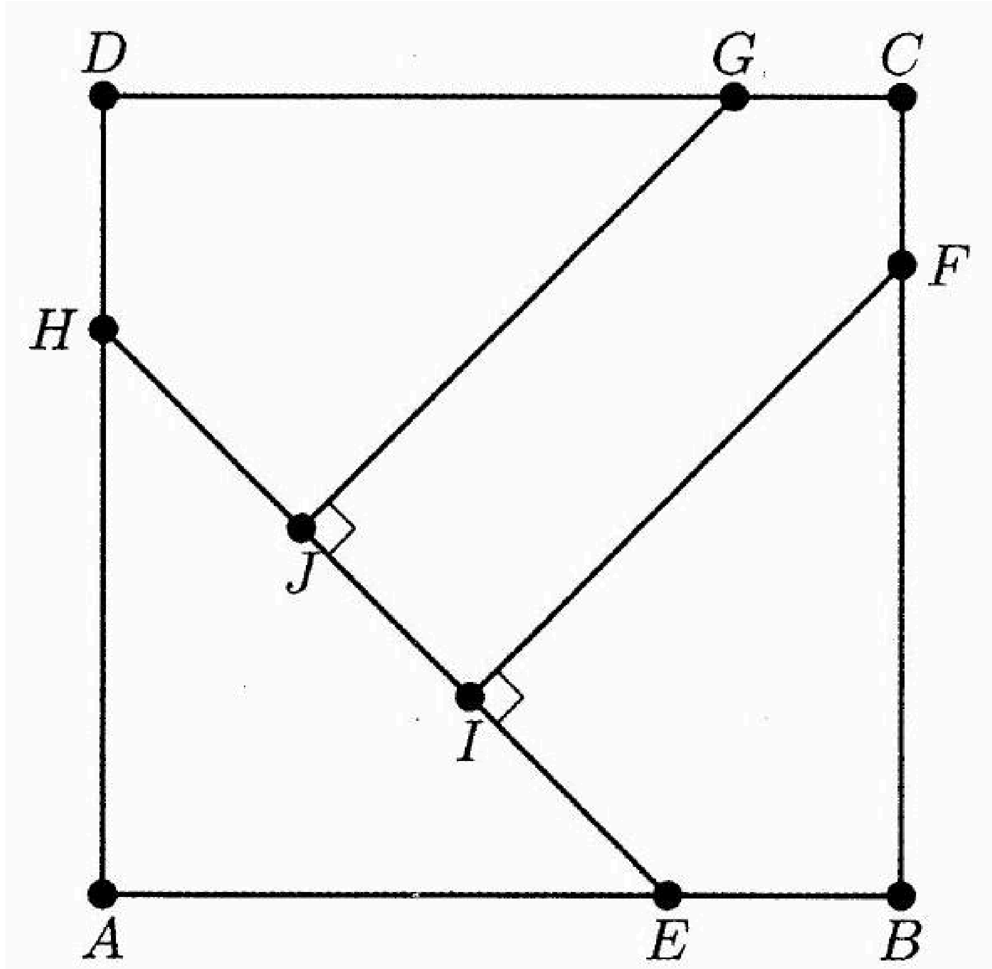
2020 AMC 10B Problem 21 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 21. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 21

Problem:
In square $ABCD$, points E and H lie on $\overline{AB}$ and $\overline{DA}$, respectively, so that $AE = AH$. Points F and G lie on $\overline{BC}$ and $\overline{CD}$, respectively, and points I and J lie on $\overline{EH}$ so that $\overline{FI} \perp \overline{EH}$ and $\overline{GJ} \perp \overline{EH}$. See the figure below. Triangle AEH, quadrilateral $BFIE$, quadrilateral $DHJG$, and pentagon $FCGJI$ each has area 1. What is FI^2?



Answer Choices:

A. $\dfrac{7}{3}$

B. $8-4\sqrt{2}$

C. $1+\sqrt{2}$

D. $\dfrac{7}{4}\sqrt{2}$

E. $2\sqrt{2}$

Solution:

Since the total area is 4, the side length of square $ABCD$ is 2. We see that since triangle HAE is a right isosceles triangle with area 1, we can determine sides HA and AE both to be $\sqrt{2}$.

Now, consider extending FB and IE until they intersect. Let the point of intersection be K. We note that EBK is also a right isosceles triangle with side $2-\sqrt{2}$ and find its area to be $3-2\sqrt{2}$.

Now, we notice that FIK is also a right isosceles triangle (because $\angle EKB = 45^\circ$) and find its area to be $\frac{1}{2}FI^2$. This is also equal to $1+3-2\sqrt{2}$ or $4-2\sqrt{2}$. Since we are looking for FI^2, we want two times this. That gives $\text{(B)}\boxed{8-4\sqrt{2}}$

OR

Draw the auxiliary line AC. Denote by M the point it intersects with HE, and by N the point it intersects with GF. Last, denote by x the segment FN, and by y the segment FI. We will find two equations for x and y, and then solve for y^2.

Since the overall area of $ABCD$ is $4 \implies AB = 2$, and $AC = 2\sqrt{2}$.

In addition the area of $\triangle AME = \frac{1}{2} \implies AM = 1$.

The two equations for x and y are then:

- Length of $AC : 1 + y + x = 2\sqrt{2} \implies x = (2\sqrt{2}-1) - y$
- Area of CMIF: $\frac{1}{2}x^2 + xy = \frac{1}{2} \implies x(x+2y) = 1.$

Substituting the first into the second, yields

$$[(2\sqrt{2}-1) - y] \cdot [(2\sqrt{2}-1) + y] = 1$$

Solving for y^2 gives $\text{(B)}\boxed{8-4\sqrt{2}}$.